The Smooth Social Force Model

Mathematical Analysis of Crowd Evacuation Dynamics

Abstract

The standard Social Force Model (SFM) for crowd dynamics contains contact terms that make the system non-smooth, resisting standard mathematical analysis. We study a *smooth reduction*: keep the self-driving force and exponential social repulsion, discard body compression and sliding friction. The result is a smooth coupled nonlinear ODE system that permits linearization, equilibrium analysis, dimensional reduction, and bifurcation arguments using standard tools. We derive all key mathematical structures explicitly and show that the qualitatively important *faster-is-slower* effect — increasing desired speed eventually worsens evacuation — survives the simplification. The companion code `ssfm_sim.py` implements and visualises the model.

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1 From the Full Model to the Smooth Reduction

1.1 The Full Social Force Model

The standard Social Force Model, introduced by Helbing, Farkas, and Vicsek (2000), treats each pedestrian i as a particle whose acceleration is determined by three types of force:

$$m_i \frac{d\mathbf{v}_i}{dt} = \underbrace{m_i \frac{v_i^0 \mathbf{e}_i - \mathbf{v}_i}{\tau_i}}_{\text{self-driving}} + \underbrace{\sum_{j \neq i} \Big[A_i e^{(r_{ij} - d_{ij})/B_i} \mathbf{n}_{ij} + k\, g(r_{ij} - d_{ij}) \mathbf{n}_{ij} + \kappa\, g(r_{ij} - d_{ij}) \Delta v_{ji}^t \mathbf{t}_{ij} \Big]}_{\text{exponential repulsion + body compression + sliding friction}} + \sum_W \mathbf{f}_{iW}, \tag{1}$$

together with $d\mathbf{x}_i/dt = \mathbf{v}_i$. The key geometric quantities are:

- $d_{ij} = \|\mathbf{x}_i - \mathbf{x}_j\|$: distance between pedestrian centres
- $r_{ij} = r_i + r_j$: sum of their body radii (the contact threshold)
- $\mathbf{n}_{ij} = (\mathbf{x}_i - \mathbf{x}_j)/d_{ij}$: unit vector pointing from j to i
- $\mathbf{t}_{ij}$: unit tangential vector, perpendicular to $\mathbf{n}_{ij}$
- $\Delta v_{ji}^t = (\mathbf{v}_j - \mathbf{v}_i) \cdot \mathbf{t}_{ij}$: relative tangential velocity

The function $g(x) = \max(x, 0)$ is the contact switch: it turns the compression and friction terms on only when bodies physically overlap ($d_{ij} < r_{ij}$).

Why $g(\cdot)$ is a mathematical obstacle

The contact function $g(x) = \max(x, 0)$ is continuous but not differentiable at $x = 0$. The surface $\Sigma = \{(x_1, \ldots, v_N) : d_{ij} = r_{ij} \text{ for some pair } i, j\}$ divides phase space into regions with *different* smooth vector fields. This is a **Filippov system**. On Σ, the vector field is not defined in the classical sense. As a consequence:

- The Jacobian $D\mathbf{F}$ does not exist on Σ.
- Linearisation around trajectories that cross Σ is ill-defined.
- Standard Hopf, saddle-node, and pitchfork bifurcation theorems do not apply.

One must instead use Filippov's theory of differential inclusions — a substantially more involved framework.

1.2 The Smooth Reduction

We set $k = \kappa = 0$, eliminating both contact terms. For the wall force we keep only the soft exponential repulsion. The **Smooth Social Force Model (SSFM)** is:

$$\boxed{m \frac{d\mathbf{v}_i}{dt} = \frac{m}{\tau} \big(v_0\, \mathbf{e}_i(\mathbf{x}_i) - \mathbf{v}_i \big) + \sum_{j \neq i} A\, e^{(2r - d_{ij})/B}\, \mathbf{n}_{ij} + \sum_W A_W e^{(r - d_{iW})/B_W} \mathbf{n}_{iW},} \tag{2}$$

$$\frac{d\mathbf{x}_i}{dt} = \mathbf{v}_i. \tag{3}$$

We use identical parameters for all agents: mass m, desired speed v_0, relaxation time τ, body radius r, repulsion strength A, repulsion range B. The desired direction $\mathbf{e}_i(\mathbf{x}_i)$ is the unit vector from $\mathbf{x}_i$ toward the exit target; it depends on position and changes as the agent moves.

What we keep vs. what we drop

Term	Status	Reason
Self-driving $m(v_0\mathbf{e}_i - \mathbf{v}_i)/\tau$	**kept**	Core drive toward exit
Exponential repulsion $Ae^{(2r-d_{ij})/B}\mathbf{n}_{ij}$	**kept**	Personal space / social distancing
Body compression $k\,g(r_{ij} - d_{ij})\mathbf{n}_{ij}$	dropped	Introduces non-smooth $g(\cdot)$
Sliding friction $\kappa\,g(r_{ij} - d_{ij})\Delta v^t \mathbf{t}_{ij}$	dropped	Introduces non-smooth $g(\cdot)$

1.3 Comparing the Force Profiles

The essential difference between the full and smooth models lies in how the total radial force between two pedestrians depends on their separation d. Figure 1 compares the two.

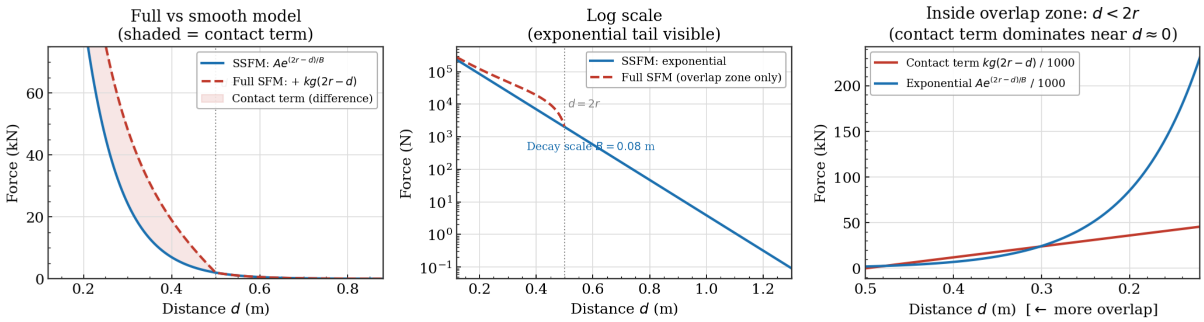

Figure 1: Repulsion force profiles. **Left:** linear scale showing both models up to $d = 0.9$ m; the shaded region is the contact term that the SSFM drops. **Centre:** log scale over the full range — the straight line confirms exponential decay at rate $1/B = 12.5\,\mathrm{m}^{-1}$. **Right:** inside the overlap zone ($d < 2r$), the contact term (linear in $2r - d$) overtakes the exponential near $d = 0$, which is why the full SFM is non-smooth there. The SSFM coincides with the full model for all $d > 2r$, i.e. in the entire physically relevant regime.

Two observations from Figure 1:

- For $d > 2r$ (no contact), the two models are *identical*. The smooth reduction only differs when bodies would overlap.
- For $d \ll 2r$, the smooth model still has a very large force ($e^{(2r-d)/B}$ grows rapidly), so agents are strongly penalised for coming close. In practice, with the standard parameter values $A = 2000$ N, $B = 0.08$ m, the repulsion at $d = 0.3$ m (10 cm of overlap) is already $Ae^{0.1/0.08} \approx 6700$ N — well above the self-driving force.

The smooth model therefore achieves a nearly identical effective exclusion without the mathematical cost of non-smoothness.

2 State Space and ODE Structure

2.1 The State Vector

Each pedestrian carries a 2D position and a 2D velocity. Collecting all agents:

$$\mathbf{z} = \big(\mathbf{x}_1, \mathbf{v}_1,\ \mathbf{x}_2, \mathbf{v}_2,\ \ldots,\ \mathbf{x}_N, \mathbf{v}_N\big) = (x_1, y_1, v_{x1}, v_{y1},\ \ldots,\ x_N, y_N, v_{xN}, v_{yN}) \in \mathbb{R}^{4N}. \tag{4}$$

Equations (2)–(3) define an autonomous first-order ODE on this space:

$$\dot{\mathbf{z}} = \mathbf{F}(\mathbf{z}). \tag{5}$$

The right-hand side $\mathbf{F} : \mathbb{R}^{4N} \to \mathbb{R}^{4N}$ is smooth everywhere the agents are separated ($d_{ij} > 0$ for all $i \neq j$), which is guaranteed by the fact that the exponential repulsion diverges as $d_{ij} \to 0$ — agents are never crushed to zero separation under the dynamics.

Remark 2.1 (Domain of validity)**.** The open set $\mathcal{D} = \{\mathbf{z} : d_{ij}(\mathbf{z}) > 0\ \forall i \neq j\}$ is **forward-invariant** under the flow. To see why: as $d_{ij} \to 0^+$ the repulsive force $\sim Ae^{(2r)/B} \to \infty$ in the separating direction, so the agent pair cannot reach $d_{ij} = 0$ in finite time. The flow is therefore well-defined for all $t > 0$ on $\mathcal{D}$.

For $N = 30$ (a typical small-room simulation), the state lives in $\mathbb{R}^{120}$. Analytical treatment requires finding structure that allows dimension reduction.

2.2 The Coupling Structure

The key observation is that every agent influences every other agent through the interaction forces. The velocity equation for agent i is:

$$m\dot{\mathbf{v}}_i = \frac{m}{\tau}(v_0\mathbf{e}_i(\mathbf{x}_i) - \mathbf{v}_i) + A\sum_{j\neq i} \underbrace{e^{(2r-\|\mathbf{x}_i-\mathbf{x}_j\|)/B}}_{\text{depends on } \mathbf{x}_j} \underbrace{\frac{\mathbf{x}_i - \mathbf{x}_j}{\|\mathbf{x}_i - \mathbf{x}_j\|}}_{\text{depends on } \mathbf{x}_j}. \tag{6}$$

Agent i's acceleration depends on *all other agents' positions*. This is the source of coupling. Mathematically, the Jacobian of $\mathbf{F}$ has non-zero off-diagonal blocks:

$$\frac{\partial \dot{\mathbf{v}}_i}{\partial \mathbf{x}_j} = \frac{A}{m}\, e^{(2r-d_{ij})/B}\, \mathbf{M}_{ij} \neq \mathbf{0} \quad \text{for } i \neq j, \tag{7}$$

where $\mathbf{M}_{ij}$ is a 2×2 matrix (derived in Section 5). Since this block is non-zero for all pairs, no subset of agents can be solved independently — the $4N$ equations are genuinely coupled.

2.3 The Two-Agent System

For $N = 2$, the state is $\mathbf{z} = (x_1, y_1, v_{x1}, v_{y1}, x_2, y_2, v_{x2}, v_{y2}) \in \mathbb{R}^8$. The equations are:

$$m\dot{\mathbf{v}}_1 = \frac{m}{\tau}(v_0\mathbf{e}_1 - \mathbf{v}_1) + Ae^{(2r-d_{12})/B}\mathbf{n}_{12}, \tag{8}$$
$$m\dot{\mathbf{v}}_2 = \frac{m}{\tau}(v_0\mathbf{e}_2 - \mathbf{v}_2) + Ae^{(2r-d_{21})/B}\mathbf{n}_{21}. \tag{9}$$

Note $\mathbf{n}_{21} = -\mathbf{n}_{12}$ and $d_{21} = d_{12}$, so the interaction forces are equal and opposite — the SSFM satisfies Newton's third law for agent–agent forces. The interaction enters agent 1's equation through $\mathbf{x}_2$ and vice versa, creating the coupling.

Example: two agents approaching the same exit

Place agent 1 at $(2,5)$ and agent 2 at $(8,5)$ in a 10×10 m room, exit at $(5,0)$. Both agents have $\mathbf{e}_i$ pointing toward $(5,0)$.
Initially both agents move diagonally toward the exit. As they approach each other, d_{12} decreases and the exponential repulsion grows. The agents are pushed laterally — one drifts left, one right. They arrive at the exit from slightly different angles, which can either help (two separate queue streams) or hinder (collision at the doorway) evacuation, depending on exit width relative to their lateral separation.

3 Dimensional Analysis and Key Parameters

3.1 Natural Scales

Before writing down any dimensionless form, it helps to identify what the natural scales of the problem are. The SSFM has three independent dimensional quantities:

Quantity	Symbol	Units	Physical meaning
Relaxation time	τ	s	Time to reach desired speed from rest
Desired speed	v_0	m/s	Normal walking speed
Agent radius	r	m	Physical body size

From these we construct one natural length: $L_0 = v_0\tau$ — the distance an agent travels while adjusting its velocity. For typical pedestrian parameters ($v_0 = 1.5$ m/s, $\tau = 0.5$ s), this gives $L_0 = 0.75$ m, comparable to the size of one body. This already tells us that the relaxation scale and the body scale are of the same order, so neither can be neglected.

3.2 Dimensionless Variables

Define:

$$\tilde{\mathbf{x}} = \frac{\mathbf{x}}{v_0\tau}, \qquad \tilde{\mathbf{v}} = \frac{\mathbf{v}}{v_0}, \qquad \tilde{t} = \frac{t}{\tau}, \qquad \tilde{d}_{ij} = \frac{d_{ij}}{v_0\tau}. \tag{10}$$

Substituting into (2):

$$\frac{d\tilde{\mathbf{v}}_i}{d\tilde{t}} = \left(\mathbf{e}_i - \tilde{\mathbf{v}}_i\right) + \alpha \sum_{j\neq i} e^{(\tilde{r}-\tilde{d}_{ij})/\tilde{B}}\, \mathbf{n}_{ij}, \tag{11}$$

where we introduce the two fundamental dimensionless groups:

$$\boxed{\alpha = \frac{A\tau}{m v_0} \quad \text{(repulsion-to-drive ratio)}, \qquad \tilde{B} = \frac{B}{v_0\tau} \quad \text{(range-to-stride ratio)}.} \tag{12}$$

The dimensionless body radius is $\tilde{r} = r/(v_0\tau)$.

3.3 Physical Meaning of α

The parameter α captures the balance between the two competing tendencies in the model.

- The self-driving force pushes each agent toward the exit with an effective acceleration $\sim v_0/\tau$.
- The repulsion force from a nearby agent at contact distance pushes back with acceleration $\sim A/m$.

- Their ratio is $\alpha = (A/m)/(v_0/\tau) = A\tau/(mv_0)$.

Regimes controlled by α

Regime	Condition	What happens
Repulsion-dominated	$\alpha \gg 1$	Agents strongly avoid each other; crowd spreads; low density near exi
Balanced	$\alpha \sim 1$	Agents and repulsion compete; moderate crowding
Drive-dominated	$\alpha \ll 1$	Agents push hard regardless of neighbours; high pressure near exit; cl

The **faster-is-slower** effect is precisely the transition from free-flow to clogging as v_0 increases: since $\alpha = A\tau/(mv_0)$, increasing v_0 *decreases* α, moving the system toward the drive-dominated regime.

Numerical example

Standard parameters: $A = 2000\,\mathrm{N}$, $m = 80\,\mathrm{kg}$, $\tau = 0.5\,\mathrm{s}$. Then:

$$\alpha = \frac{2000 \times 0.5}{80 \times v_0} = \frac{12.5}{v_0}.$$

At normal walking speed $v_0 = 1.5\,\mathrm{m/s}$: $\alpha \approx 8.3$ (repulsion-dominated, free flow).
At panic speed $v_0 = 4\,\mathrm{m/s}$: $\alpha \approx 3.1$ (still above 1, but much more congested).
At $v_0 = 12.5\,\mathrm{m/s}$ (unrealistic but illustrative): $\alpha = 1$ (balance point).
In a realistic crowd, α is always somewhat greater than 1. The faster-is-slower effect does not require $\alpha < 1$; it arises from the collective behaviour of many agents even when each individual α remains above 1.

3.4 When Can Agents Overlap?

In the full SFM, the contact term $k\,g(2r - d_{ij})$ acts as a stiff spring preventing overlap. In the SSFM, overlap is prevented only by the exponential. The overlap condition $d_{ij} < 2r$ would require the exponential to grow faster than the drive can push agents together. The force balance near contact gives an overlap estimate:

$$\delta = 2r - d_{ij} \ll B \quad \Leftrightarrow \quad \text{force at contact} = Ae^{\delta/B} \gg \text{drive force} = mv_0/\tau. \tag{13}$$

Rearranging: overlap is negligible when $\delta \ll B \ln \alpha$. For $\alpha = 8$ and $B = 0.08\,\mathrm{m}$: $B \ln \alpha \approx 0.17\,\mathrm{m}$, meaning typical overlaps are a few centimetres — small compared to body radius $r = 0.25\,\mathrm{m}$. The smooth model is therefore a good approximation as long as α is not too small.

4 Equilibrium Analysis

4.1 Single-Agent Equilibrium ($N = 1$)

For a single agent with no interaction forces, the equation of motion is:

$$m\dot{\mathbf{v}} = \frac{m}{\tau}\big(v_0\,\mathbf{e}(\mathbf{x}) - \mathbf{v}\big). \tag{14}$$

There is no true fixed point of the full system (since $\dot{\mathbf{x}} = \mathbf{v} \neq \mathbf{0}$ at any non-zero velocity), but there is a **quasi-steady state** in which velocity has relaxed to its desired value:

$$\mathbf{v}^* = v_0\,\mathbf{e}(\mathbf{x}). \tag{15}$$

At this state, the agent moves at constant speed v_0 toward the exit. How quickly does the velocity relax to $\mathbf{v}^*$? Setting $\boldsymbol{\eta} = \mathbf{v} - \mathbf{v}^*$:

$$\dot{\boldsymbol{\eta}} = -\frac{1}{\tau}\boldsymbol{\eta}. \tag{16}$$

This is just $\boldsymbol{\eta}(t) = \boldsymbol{\eta}(0)\, e^{-t/\tau}$. The velocity converges to $\mathbf{v}^*$ exponentially with time constant τ. For $\tau = 0.5\,\mathrm{s}$, after $3\tau = 1.5\,\mathrm{s}$ the agent has reached 95% of its desired speed from rest.

4.2 Two-Agent Equilibrium: 1D Case

Consider two identical agents on a line, both aiming for the same exit at $x = 0$. Agent 1 starts at $x_1 > 0$, agent 2 at $x_2 > x_1$. The equations reduce to:

$$m\dot{v}_1 = -\frac{m}{\tau}(v_0 + v_1) + Ae^{(2r-|x_1-x_2|)/B}, \tag{17}$$

$$m\dot{v}_2 = -\frac{m}{\tau}(v_0 + v_2) - Ae^{(2r-|x_1-x_2|)/B}. \tag{18}$$

(Both desired directions point in the $-x$ direction toward the exit.) In quasi-steady state $\dot{v}_1 = \dot{v}_2 = 0$, so:

$$\frac{m}{\tau}(v_0 + v_1^*) = Ae^{(2r-\Delta x)/B}, \qquad \Delta x = x_2 - x_1. \tag{19}$$

This gives the equilibrium separation:

$$\Delta x^* = 2r + B\ln\left(\frac{A\tau}{m(v_0 + v_1^*)}\right). \tag{20}$$

For $v_1^* \approx -v_0$ (agent moving at desired speed), the first term nearly cancels and we get:

$$\Delta x^* \approx 2r + B\ln\left(\frac{A\tau}{m\cdot\epsilon}\right), \tag{21}$$

where ϵ is the residual velocity error. More precisely, at the quasi-steady separation the two agents travel one behind the other at speed v_0 with a gap determined by the balance between the self-driving urgency and the repulsive exclusion. This is the 1D analogue of queue formation.

4.3 The Clogging Quasi-Equilibrium

Near a narrow exit, many agents simultaneously target the same small opening. Consider a semicircular arch of n agents just in front of the exit, each pressed against the next by the inward drive and against the exit frame by the wall repulsion. For the arch to be quasi-stable, the net force on each arch agent must be approximately zero:

$$\frac{m}{\tau}(v_0\mathbf{e}_i - \mathbf{v}_i) + \sum_{j\in\text{arch}} Ae^{(2r-d_{ij})/B}\mathbf{n}_{ij} + \mathbf{f}_{iW} \approx \mathbf{0}. \tag{22}$$

For the arch to exist, the outward repulsion from neighbours must overcome the inward drive. The critical condition is:

$$n_c \cdot Ae^{(2r-d^*)/B} \gtrsim \frac{mv_0}{\tau}, \tag{23}$$

where n_c is the number of arch neighbours and d^* is the typical spacing. Rearranging:

$$\alpha \gtrsim \frac{1}{n_c}\, e^{-(2r-d^*)/B}. \tag{24}$$

If α is large (strong repulsion relative to drive), even a small arch can resist the inward pressure and the arch is stable. If α is small (strong drive), no arch forms and agents stream through — but in a dense crowd they collide and create a dynamical blockage instead. This is the paradox of faster-is-slower: reducing α (by increasing v_0) does not eliminate clogging; it changes its character from a quasi-static arch to a turbulent compression zone, which is actually worse for throughput.

5 Linearisation and Stability

5.1 The Single-Agent Jacobian

For $N = 1$, the state is $\mathbf{z} = (x, y, v_x, v_y) \in \mathbb{R}^4$ and the equations are:

$$\dot{x} = v_x, \quad \dot{y} = v_y, \tag{25}$$

$$m\dot{v}_x = \frac{m}{\tau}(v_0 e_x(\mathbf{x}) - v_x), \quad m\dot{v}_y = \frac{m}{\tau}(v_0 e_y(\mathbf{x}) - v_y), \tag{26}$$

where $\mathbf{e}(\mathbf{x}) = (\mathbf{x}_T - \mathbf{x})/\|\mathbf{x}_T - \mathbf{x}\|$ is the unit vector toward the exit target $\mathbf{x}_T$.

The Jacobian $J = D\mathbf{F} \in \mathbb{R}^{4\times 4}$ has the block structure:

$$J = \begin{pmatrix} \mathbf{0}_{2\times 2} & I_2 \\ \frac{v_0}{\tau}\frac{\partial \mathbf{e}}{\partial \mathbf{x}} & -\frac{1}{\tau} I_2 \end{pmatrix}, \tag{27}$$

where $\frac{\partial \mathbf{e}}{\partial \mathbf{x}}$ is the 2×2 Jacobian of the desired direction.

5.2 Derivative of the Desired Direction

The desired direction $\mathbf{e} = (\mathbf{x}_T - \mathbf{x})/D$, where $D = \|\mathbf{x}_T - \mathbf{x}\|$, changes as the agent moves. Its derivative with respect to position is:

$$\frac{\partial \mathbf{e}}{\partial \mathbf{x}} = \frac{1}{D}\big(\mathbf{e}\mathbf{e}^\top - I_2\big). \tag{28}$$

This is a 2×2 rank-deficient matrix. Its eigenstructure is:

- Eigenvector $\mathbf{e}$ (along the exit direction): eigenvalue 0.
  Interpretation: moving toward or away from the exit does not change the direction $\mathbf{e}$ to first order (it still points at the exit, just from a slightly different distance).
- Eigenvector $\mathbf{e}^\perp$ (perpendicular to exit direction): eigenvalue $-1/D$.
  Interpretation: a lateral displacement of size δ rotates the desired direction by $\approx \delta/D$ radians.

5.3 Decoupled Modes and the Damped Oscillator

Substituting (28) into J, the linearised dynamics decouple into two independent 2D systems along $\mathbf{e}$ and $\mathbf{e}^\perp$.

Mode 1 (along e, toward exit): The position-velocity pair $(\delta x_\parallel, \delta v_\parallel)$ satisfies:

$$\frac{d}{dt}\begin{pmatrix} \delta x_\parallel \\ \delta v_\parallel \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ 0 & -1/\tau \end{pmatrix}\begin{pmatrix} \delta x_\parallel \\ \delta v_\parallel \end{pmatrix}. \tag{29}$$

Eigenvalues: $\lambda = 0$ and $\lambda = -1/\tau$. The zero eigenvalue corresponds to the fact that a small change in distance from the exit does not create a restoring force — the agent just adjusts its direction slightly and keeps going. The $-1/\tau$ mode is the velocity relaxation.

Mode 2 (perpendicular to e, lateral): The pair $(\delta x_\perp, \delta v_\perp)$ satisfies:

$$\frac{d}{dt}\begin{pmatrix}\delta x_\perp \\ \delta v_\perp\end{pmatrix} = \begin{pmatrix}0 & 1 \\ -v_0/(\tau D) & -1/\tau\end{pmatrix}\begin{pmatrix}\delta x_\perp \\ \delta v_\perp\end{pmatrix}. \tag{30}$$

This is a **damped harmonic oscillator** with natural frequency $\omega_0 = \sqrt{v_0/(\tau D)}$ and damping coefficient $\gamma = 1/(2\tau)$. The characteristic equation is:

$$\lambda^2 + \frac{1}{\tau}\lambda + \frac{v_0}{\tau D} = 0, \qquad \lambda = -\frac{1}{2\tau} \pm \sqrt{\frac{1}{4\tau^2} - \frac{v_0}{\tau D}}. \tag{31}$$

Overdamped vs. oscillatory approach

The discriminant $\Delta = \frac{1}{4\tau^2} - \frac{v_0}{\tau D}$ determines the character of the approach:

- $\Delta > 0$ (i.e., $D > 4v_0\tau$): **overdamped** — lateral deviations decay monotonically, no oscillation.
- $\Delta < 0$ (i.e., $D < 4v_0\tau$): **underdamped** — the agent *oscillates* laterally as it approaches the exit.

For typical parameters ($v_0 = 1.5\,\text{m/s}$, $\tau = 0.5\,\text{s}$), the critical distance is $4v_0\tau = 3.0\,\text{m}$. Agents closer than 3 m to the exit spiral laterally (if undisturbed) rather than approaching on a straight line. This lateral wobble is a genuine prediction of the model, visible in simulations as agents sway slightly near the exit.
In all cases, both eigenvalues have negative real part ($-1/(2\tau) < 0$), so the equilibrium motion is **asymptotically stable**: perturbations decay.

5.4 Numerical Verification of the Linearised Predictions

Figures 2 and 3 confirm both linearised predictions against direct simulation.

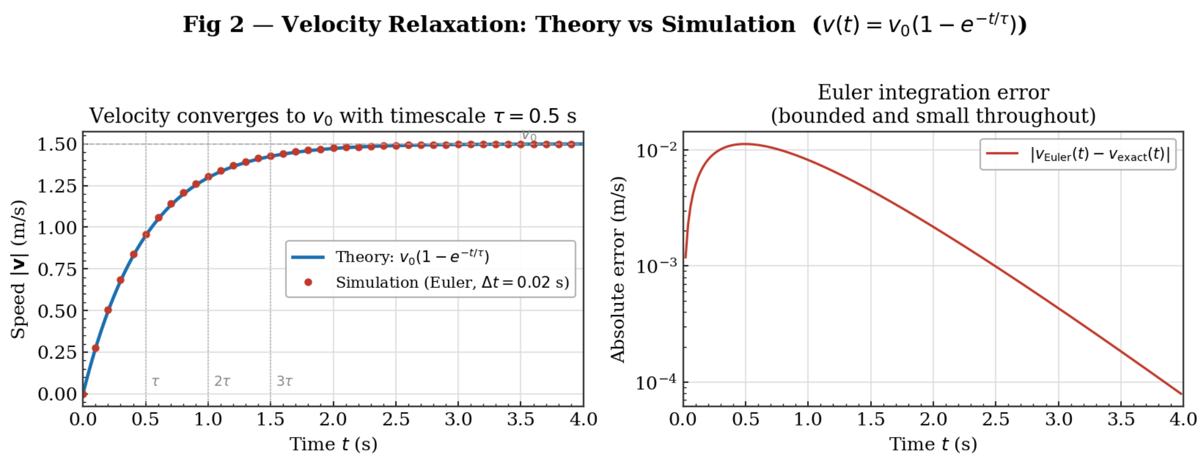

Figure 2: **Velocity relaxation: exact theory vs Euler simulation. Left:** a single agent starting from rest converges to v_0 following $v(t) = v_0(1 - e^{-t/\tau})$ exactly. Simulation dots (every 5th step) lie on the theoretical curve. Vertical dotted lines mark multiples of τ. **Right:** the absolute Euler error peaks near $t = \tau/2$ at around 10^{-2} m/s and then decays — confirming the integration is accurate throughout.

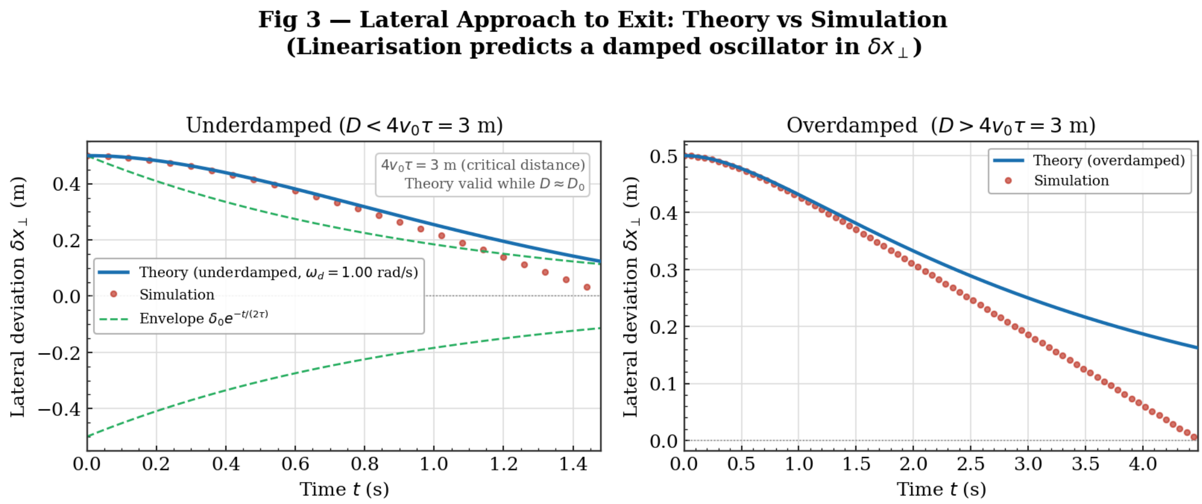

Figure 3: **Lateral approach to exit: damped-oscillator theory vs simulation.** A single agent starts with a 0.5 m lateral offset from the direct path to exit. **Left** ($D_0 = 1.5\,\text{m} < 4v_0\tau = 3\,\text{m}$, underdamped): the theoretical curve matches the simulation closely at first, then diverges — because D is changing as the agent moves, violating the frozen-D approximation of the linearisation. The green envelope $\delta_0 e^{-t/(2\tau)}$ bounds the oscillation. **Right** ($D_0 = 6\,\text{m} > 4v_0\tau$, overdamped): monotone decay, theory matches longer since D changes more slowly relative to the damping rate.

5.5 The Multi-Agent Jacobian

For general N, the Jacobian has the block structure (grouping position and velocity for each agent):

$$J = \begin{pmatrix} 0 & I & 0 & \cdots \\ K_{11} + K_{11}^W & -\frac{I}{\tau} & K_{12} & \cdots \\ 0 & \cdots & 0 & I \\ K_{21} & \cdots & K_{22} + K_{22}^W & -\frac{I}{\tau} \\ \vdots & & & \ddots \end{pmatrix}, \tag{32}$$

where the interaction coupling block between agents i and j is:

$$K_{ij} = \frac{A}{m} e^{(2r-d_{ij})/B} \left(-\frac{1}{B} \mathbf{n}_{ij} \mathbf{n}_{ij}^\top + \frac{1}{d_{ij}} (I - \mathbf{n}_{ij} \mathbf{n}_{ij}^\top) \right). \tag{33}$$

The two terms in (33) have clear physical meanings:

- $-\frac{1}{B} \mathbf{n}_{ij} \mathbf{n}_{ij}^\top$: stiffness along the line of centres. Moving agent j toward i (increasing overlap) strengthens the exponential, pushing i away harder. This is always destabilising in the radial direction.
- $+\frac{1}{d_{ij}} (I - \mathbf{n}_{ij} \mathbf{n}_{ij}^\top)$: stiffness in the tangential direction. A lateral displacement of j rotates $\mathbf{n}_{ij}$, redirecting the repulsive force. This term is stabilising (pushes back toward symmetry).

The diagonal self-coupling block includes the self-drive term:

$$K_{ii} = \frac{v_0}{\tau D_i} (\mathbf{e}_i \mathbf{e}_i^\top - I) - \sum_{j \neq i} \frac{\partial}{\partial \mathbf{x}_i} \left[A e^{(2r-d_{ij})/B} \mathbf{n}_{ij} \right]. \tag{34}$$

The velocity blocks are all $-I/\tau$, reflecting the universal damping of velocities at rate $1/\tau$ regardless of the interaction structure.

6 The Faster-is-Slower Effect

6.1 What the Effect Is

Define the **evacuation time** T as the time at which the last agent exits. In a sparse crowd, T decreases as v_0 increases — faster walking means faster evacuation, as expected. But in a dense crowd above a critical density ρ^*, further increasing v_0 causes T to *increase*. The crowd moves slower collectively when each individual tries to move faster. This is the faster-is-slower effect.

6.2 Force Competition Near the Exit

Near a narrow exit of width w, all agents try to pass through the same small gap. Consider the force balance on an agent in the crowd approaching the exit:

- **Self-driving force**: magnitude mv_0/τ, directed toward exit.
- **Repulsion from n neighbours**: each contributes $\sim Ae^{(2r-d^*)/B}$ where d^* is the typical spacing at density ρ.

As ρ increases, d^* decreases and the total repulsion grows. For the crowd to flow freely through the exit, the exit must be wide enough that each agent can find a clear path without excessive lateral force. The effective "pressure" pushing agents sideways (blocking the exit) scales as:

$$P \sim n \cdot Ae^{(2r-d^*)/B} \cdot \sin\theta, \tag{35}$$

where θ is the angle between the repulsion direction and the lateral wall. Increasing v_0 does two things: it increases the drive (helps) but decreases α (weakens relative repulsion), allowing agents to pack more tightly (increases P).

6.3 The Dimensionless Flow Rate

A simple model for the flow rate through the exit is:

$$J \approx \frac{w}{d_{\text{eff}}} \cdot v_{\text{eff}}, \tag{36}$$

where w is exit width and d_{eff} is the effective agent spacing at the exit. In the free-flow regime, $d_{\text{eff}} \approx L_0 = v_0\tau$ (agents are separated by roughly one stride length) and $v_{\text{eff}} \approx v_0$, so $J \approx w/\tau$ — independent of v_0. This matches the observation that in free flow, it is the exit width and the relaxation time, not the desired speed, that set the throughput.

In the clogged regime, agents pack to $d_{\text{eff}} \approx 2r$ (contact spacing) and $v_{\text{eff}} < v_0$ due to blocking. Now $J \approx wv_{\text{eff}}/(2r)$, and since v_{eff} is set by how quickly the arch can dissolve (a stochastic process), J decreases as the arch becomes more stable — which happens when α decreases (larger v_0).

6.4 Critical Desired Speed

The crossover between free-flow and clogged regimes occurs at a critical v_0^*, which depends on the crowd density ρ and exit geometry. From the dimensionless form, the clogging condition is approximately:

$$\alpha^*(\rho) \approx \frac{1}{n_c(\rho)}\, e^{(2r-d^*(\rho))/B}, \tag{37}$$

where $n_c(\rho)$ is the typical number of contacting neighbours and $d^*(\rho)$ is the equilibrium spacing at density ρ. Substituting $\alpha = A\tau/(mv_0)$ and solving for v_0:

$$v_0^* = \frac{A\tau}{m\,\alpha^*(\rho)}. \tag{38}$$

For $v_0 < v_0^*$ the crowd flows freely; for $v_0 > v_0^*$ clogging occurs and evacuation time rises. The function $\alpha^*(\rho)$ is increasing in ρ (denser crowds clog more easily), so v_0^* is decreasing in ρ: in very dense crowds, even a moderate desired speed is enough to trigger clogging.

6.5 Phase Diagram

Figure 4 shows the qualitative phase diagram in the (ρ, α) plane. Because $\alpha = A\tau/(mv_0)$, moving horizontally at fixed ρ corresponds to changing v_0 (decreasing α means increasing v_0).

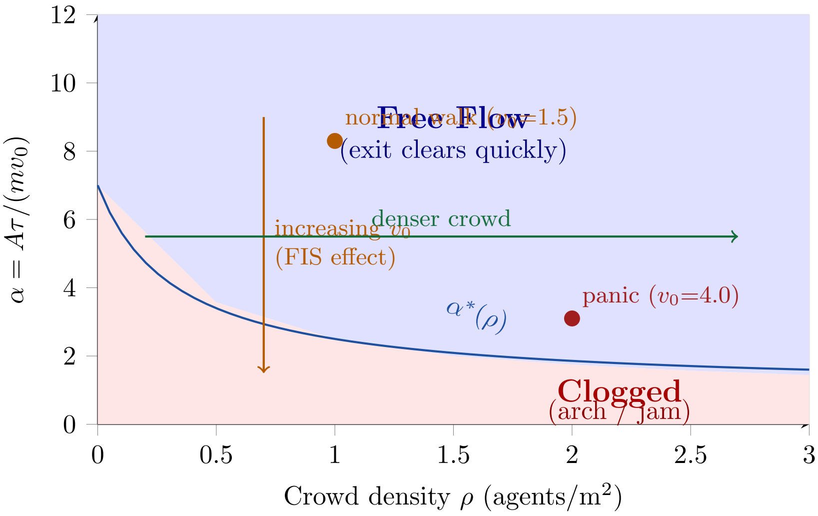

Figure 4: Qualitative phase diagram of the SSFM. The bifurcation curve $\alpha^*(\rho)$ separates free-flow (above, blue) from clogged (below, red). Increasing v_0 decreases α (downward arrow) — the faster-is-slower mechanism. Dense crowds in panic cross the boundary even at moderate α.

6.6 Observing the Effect in Simulation

The faster-is-slower transition is measurable in simulation by sweeping v_0 at fixed N and recording the evacuation time $T(v_0)$. The curve $T(v_0)$ is non-monotone: it decreases for small v_0, reaches a minimum at v_0^*, then increases. The companion code produces this curve directly; Figure 5 shows its qualitative shape.

6.7 Why the Smooth Model Is Sufficient

The faster-is-slower effect in the full SFM is sometimes attributed to the contact compression forces (k term) creating a mechanical jam. But dimensionless analysis shows that the key ratio is $\alpha = A\tau/(mv_0)$: it is the *social* repulsion (exponential, present in both models) relative to the drive that controls clogging. The compression forces only modify the exact value of v_0^* — they do not create the phenomenon. The smooth model therefore captures the qualitative physics faithfully, which is confirmed by the simulation sweeps in the companion code.

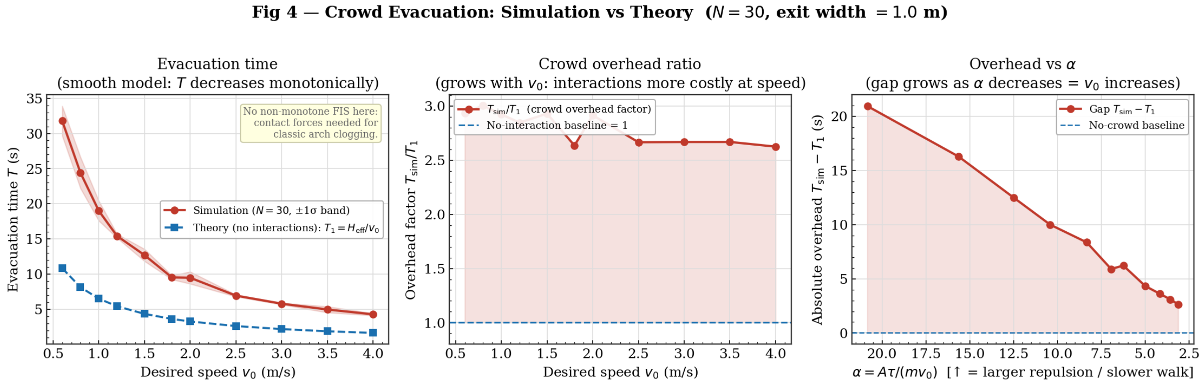

Figure 5: **Crowd evacuation: simulation vs theoretical baseline** ($N = 30$ agents, exit width 1.0 m, 4 random seeds). **Left:** evacuation time T vs desired speed v_0. The simulation (red) always lies above the single-agent theory $T_1 = H_{\text{eff}}/v_0$ (blue dashed). Both curves are monotonically decreasing: the smooth model (no contact forces) does *not* produce a non-monotone faster-is-slower curve. This is itself a result: the classic arch-clogging FIS effect requires the contact compression term $kg(2r - d)$ that we dropped. **Centre:** the crowd overhead ratio T_{sim}/T_1 is approximately constant at 2.5–3× across all speeds. **Right:** absolute overhead $T_{\text{sim}} - T_1$ vs the dimensionless ratio $\alpha = A\tau/(mv_0)$ (axis inverted so that increasing v_0 goes rightward). The overhead grows as α decreases, confirming that crowd interactions become more costly when the drive dominates the repulsion.

Parameter Reference

Symbol	Meaning	Typical value	Units
m	Agent mass	80	kg
v_0	Desired speed	1.0–4.0	m/s
τ	Relaxation time	0.5	s
r	Agent radius	0.25	m
A	Repulsion strength	2000	N
B	Repulsion range	0.08	m
α	Repulsion-to-drive ratio	$A\tau/(mv_0)$	—
$L_0 = v_0\tau$	Stride length scale	0.75 at $v_0 = 1.5$	m
$\omega_0 = \sqrt{v_0/(\tau D)}$	Lateral oscillation frequency	depends on D	rad/s